

POKHARA UNIVERSITY

Level: Bachelor

Semester –Spring

Year:2020

Program: BE

Full Marks: 70

Course: Microprocessor and Assembly Language Programming

Pass Marks:31.5

Time: 2 hrs.

Candidates are required to answer in their own words as far as practicable.

The figures in the margin indicate full marks.

Group - A: Attempt all questions (5×10=50)

- Q. N. 1 Explain the architecture of a 8-bit microprocessor clarifying the execution of instruction in term of a block diagram.

OR

What are the architectural and functional advancement in technology that made the modern days processors more powerful and faster than the previous generation? Express in your own words

- Q. N. 2 WAP in 8085 :

- i) To find the largest number in an array of 'n' numbers stored in memory locations starting from CODOH. Assume that 'n' is the last two digit of your exam roll number.
- ii) To swap the contents of memory location using indirect addressing. Memory locations [5000] =AA and [6000] =BB. Explain each step in detail.

- Q. N. 3 Suppose you were doing some project using 8085 microprocessor.

- i. During your program execution in 8085, suppose INTR signal is raised high, explain what happens after this and how it is solved?
- ii. If the data transfer rate between the processor and peripheral device needs to be improved, Which device would you use? Explain its operation along with its diagram.

- Q. N. 4 What do you mean by address decoding? Differentiate between I/O mapped I/O and memory mapped I/O. Design an address decoding circuit to interface two RAM block each of size 4KB starting at address 5000H.

- Q. N. 5 Write an assembly language program for 8086 microprocessors to check how many times the last digit of your symbol number is repeated in your symbol number and display the result. [For example if your symbol number is 321123, the last digit '3' is repeated 2 times, so your program should display the result: 2].

Group - B: (1×20=20)

- Q. N. 6 You are given an instruction ADI 25H. Draw and explain the timing diagram for this instruction. Explain the instruction in term of its byte size and addressing mode. Show the status of all the FLAGS after the execution of this instruction. Assume the content of the Accumulator to be the last two digit of your exam roll number.